

Challenge: Gap Crossing

Points: 300 • Estimated Time: 60 Minutes

Challenge Scope

The robot must safely traverse a 10cm wide gap or bridge layout without falling off the elevated track.

Instructions & Engineering Hints

- **Objective:** Your goal is to program and configure your robot to successfully cross a 10-cm wide gap on an elevated track without falling off or getting stuck.
- **Structural Considerations:** Think about the length and weight distribution of your robot. If a robot is too short, its wheels might drop directly into the gap. Consider how adding structural extensions, skids, or passive balancing rollers in the front or rear could bridge the physical void.
- **Sensor Strategy:** Decide whether you will rely strictly on autonomous navigation using sensors (like distance sensors to detect the edge of the platform) or if you will design a mechanical bridge-crossing mechanism that forces the wheels to stay safely supported.
- **Testing & Iteration:** Test your approach incrementally. Run trials at reduced motor speeds to observe exactly where your robot loses traction or balance before attempting full-speed crossings.